Assignment 1
Due on September 27, 2021, at 8am

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ID NUMBER:	U2210251	SECTION:	004

SPECIAL INSTRUCTIONS

– Let n be the LAST THREE digits of your ID. For example, if your ID is U1501931, then $n = 931$.
– Using YOUR value of n, solve the differential equations below.
– YOU WILL GET 0 IF THE EQUATIONS ARE SOLVED FOR DIFFERENT OR GENERAL n.
– YOU SHOULD SUBMIT YOUR WORK WITH THIS FILLED FORM ATTACHED.

1. Solve the initial value problem:

$$\frac{dx}{dt} = (n+1)tx^n, \quad x(0) = 2n.$$

2. Find the general solution to the equation:

$$x\frac{dy}{dx} - y = x^2e^{nx}.$$

3. Solve the initial value problems:

$$(1 + (n+1)x^n \sin y)dx + (x^{n+1} \cos y)dy = 0, \quad x(0) = 1.$$

4. Find the general solution to the equations:

$$\frac{dy}{dx} - y = e^{2x}y^n.$$

① U2210251

$$\frac{dx}{dt} = (251+1)\, t\, x^{251}$$

$$\frac{dx}{x^{251}} = (252t)\, dt$$

$$\int \frac{dx}{x^{251}} = \int (252)\, t\, dt$$

$$-\frac{1}{250 x^{250}} = 126t^2 + C$$

$$250 x^{250} = -\frac{1}{126t^2 + C}$$

$$x = \sqrt[250]{-\frac{1}{250(126t^2 + C)}}$$

$$x(0) = \sqrt[250]{-\frac{1}{250(126 \cdot 0^2 + C)}} = 502$$

$$-\frac{1}{250 \cdot C} = 502^{250}$$

$$C = -\frac{1}{250 \cdot 502^{250}}$$

$$C \approx -2{,}67 \cdot 10^{-678}$$

$$\cancel{\frac{dy}{dx} - y = e^{2x} x^n}$$

② $$x \frac{dy}{dx} - y = x^2 e^{251x} \quad | \cdot (-1)$$

$$-x \frac{dy}{dx} + y = -x^2 e^{251x} \quad | (-1)$$

$$\frac{dy}{dx} - \frac{y}{x} = x e^{251x}$$

$$\mu(x) = e^{\int -\frac{1}{x} dx} = \frac{1}{x}$$

$$\frac{1}{x} \cdot \frac{dy}{dx} - \frac{y}{x^2} = e^{251x}$$

as $\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$ and $y' = \frac{dy}{dx}$

$$\left(\frac{y}{x}\right)' = e^{251x} \Rightarrow \frac{y}{x} = \int e^{251x}$$

$$\frac{y}{x} = \frac{e^{251x}}{251} \Rightarrow y = \frac{x e^{251x}}{251}$$

(3) $\underbrace{(1+(251+1)x^{251}\sin y)}_{M}dx+\underbrace{(x^{252}\cos y)}_{N}dy=0$; $x(0)=1$

is $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$? $\Rightarrow 252x^{251}\cos y=252x^{251}\cos y$, yes, exact

$$\begin{cases}\frac{\partial F}{\partial x}=1+252x^{251}\sin y\\ \frac{\partial F}{\partial y}=x^{252}\cos y \Rightarrow F=x^{252}\sin y+f(x)\Rightarrow \frac{\partial F}{\partial x}=\cancel{252x^{251}\sin y}+\end{cases}$$

$+f'(x)=1+\cancel{252x^{251}\sin y}\Rightarrow f'(x)=1\Rightarrow f(x)=x$

$F=x^{252}\sin y+x=C$ $\Rightarrow$ when $x(0)=1\Rightarrow 1^{252}\cdot\sin 0+1=C\Rightarrow C=1$

(4) $\frac{dy}{dx}-y=e^{2x}y^{251}$

$z(x)=y^{-250}\Rightarrow y=z^{-\frac{1}{250}}$

$(z^{-\frac{1}{250}})'\frac{dz}{dx}-z^{-\frac{1}{250}}=e^{2x}z^{-\frac{251}{250}}$

$-\frac{1}{250}z^{-\frac{251}{250}}\frac{dz}{dx}-z^{-\frac{1}{250}}=e^{2x}z^{-\frac{251}{250}}\ \Big|\cdot z^{\frac{251}{250}}$

$-\frac{1}{250}\cdot\frac{dz}{dx}-z=e^{2x}$

$\frac{1}{250}\cdot\frac{dz}{dx}+z=-e^{2x}$

$\cancel{as\ (\frac{1}{250})'}=\frac{dz}{dx}+250z=-250e^{2x}$

$\mu(x)=e^{\int 250dx}=e^{250x}$

$e^{250x}\frac{dz}{dx}+e^{250x}\cdot 250z=-250e^{252x}$

$(e^{250x}\cdot z)'=-250e^{252x}$

$e^{250x}\cdot z=-125e^{252x}+C$

$z=-\frac{125}{126}\cdot e^{2x}$

$y=\sqrt[-250]{-\frac{125}{126}e^{2x}}$